ActInf GuestStream #019.1 ~ Marco Facchin: Extended Predictive Minds: d

GuestStream | Marco Facchin: Extended Predictive Minds: do Markov Blankets Matter? | 1:24:27

Hello, everyone. Welcome to ActInflab guest stream number 19.1. It's March 25th, 2022.
And we are here with Marco Fackin. And we are going to be discussing the recent paper,
Extended Predictive Minds, Do Markov Blankets Matter? So if you're watching live, please feel
free to put comments or questions in the live chat and hopefully we can address them. And Marco,
really appreciate you joining. Looking forward to hearing about this paper and your lines of
research. So it would be awesome to hear an introduction and take it away. Okay. Hi,
everyone. So I'm Marco Fackin. I actually am a PhD candidate in Italy. And the paper basically
is an attempt to understand what was going on in the active inference literature with the externalism,
internalism debate from a philosophical perspective. So I studied philosophy for the most part. I'm not
super mathematical savvy. And I have basically a background in embodied extended cognition.
And I stumbled upon the papers by Hoey or Amstead and other. And I wanted to understand what was
going on with the Markov Blanket talk. And so I wrote a paper about it to understand what was going on.
And to try to suggest that maybe that much emphasis on Markov Blanket is a bit misplaced.
And yes, that's it for a general introduction for the paper, I think. And then I'm here for any
question or any doubt or anything. Okay, cool. So maybe I'll just ask some questions and we'll just see
where it goes. For somebody who was unfamiliar with extended cognitive processes or the state of the
art, maybe outside of active inference, just what is this area? How are people looking at it outside of
the active inference, FEP angle? And like, what is the contemporary state of that area?
So, okay. So the extended cognition stuff, say officially is something that spawned out of paper
by Clark and Chalmers in the early 2000s, basically. And their original idea was, as I understand it at
the beginning, the first time I was going to say, is that sometimes at least you interact with the
environment in a way such that part of cognition happens in the environment itself. So my go-to
example is using pen and paper to do math. And when you use pen and paper to do math, you write down partial
results. And that scrap of paper is working as a piece of your working memory. It's just that it's not inside
your brain. And so that was basically the original idea. And from that, there has been a rich elaboration
with people like Richard Minari suggesting that actually, DDD, we should think of extended cognitive
processes as processes that transform our cognitive capacities is not just more stuff, more memory,
more symbols, but new stuff, new capacities, new symbols, new cognitive processes that take place in the
interaction in between the subject and the environment. So we have that line of research. And then you have a more,
say, say, an active line of research that tends to stress the idea that basically all cognition takes
place in the world interacting with affordances or interacting with your surrounding environments.
But I'm not super expert on that line of research and the extended minor a bit more conservative.
So DDD punchline would be this, that sometimes stuff in the external world really is in the most literal

sense possible, a cog of the machinery that does your thinking. That is the clearest and in the most forceful way in which I'm able to put it the idea of the extended mind. And of course, sorry, once you have this idea on the table, the question is, okay, but where, when, where does it stop? Is pen and paper is in, but I don't know, are our cell phones in? Are our books in here at my left? You might not be able to see it, but there is a big library with a lot of books I read. Are all those books part of my memory? Probably not. But you need some sort of boundary. And that's you for Mark of Black, of course. So, I'll leave it back to this.

Excellent. Well, I like this, how you said it was like a cog and it's kind of like cognition. So it's right there in the word, even though that's a little bit of a wordplay. And also very interesting about the more is different. Like we're not just augmenting the computer with a bigger hard drive or more RAM. It truly can be different. And so I think maybe we could return to this idea of like, what is the space of the adjacent possible for cognitive systems, especially if they compose in these nonlinear ways. But could you maybe walk us through some of the pieces of the paper? Like what brought you to the introduction? What brought you to writing this paper on Markov blankets? And then what did you sort of weave together in the introduction? How did you clash them in the middle? And then where did you leave us at the end? So did you walk through to the paper is something not super sure I'll be able to do, satisfactory at least. So the first step was to what motivated me was to try and figure out what was going on between Andy Clark, Jacob Hoey, Ramstad and Markov blankets, of course. And what I wanted to argue basically was something like this. That even if you set aside DDD works by Yelle Brunneberg and others or Menari and Gillette that tend to give you this sort of message like we have a perfectly fine mathematical concept of Markov blankets and then a strange concept of Written blankets that is a bit mathematical and a bit ontological and we are not really sure what it is. Even if you leave this sort of worries aside, what I wanted to say is that the way in which Markov blankets are used for when it comes to discussing the extended mine stuff is very limited to the specific application. But for the extended mine, the usage of Markov blankets is not really useful and does nothing of particular value and tends to give a big headache to the people that are not into active inference.

So I would recommend not using them. The main arguments, I think there is no main argument. I think I have three arguments that are, well, first, in order to find a Markov blanket, it seems that you must already know what is inside. If you think about a graph, there is no way in which I can give you a graph and just ask you, OK, please show me where is the Markov blanket in there. I need to tell you about which node I'm interested in. So you must already know what is inside. And when it comes to the mind, the cognitive machinery, knowing what is inside is knowing which are the codes, which presumably include the brain plus maybe stuff in the environment, pen and paper and whatever. And if you already know that, which you must already know in order to find the

the relevant Markov blanket, then intuitively you already know whether the mind extends or not.
And if you already know whether the mind extends or not, you know whether it extends or not.
There is no new piece of information to be gained by thinking about whether mind extends or not in terms of Markov blanket.
That is the first argument. The second one, which is a bit messy.
Can I just pause there and make a comment on the first one?
Yeah.
OK, awesome. So though the Markov blanket, if only it were like the blankets on our bed and you could pick it up and it was a blanket, whether it was inside the box or around the box or whatever, that's not really how they are defined. It's really like a co-partitioning of internal and external states with respect to a blanket or a blanket with respect to some other internal and external states. So the blanket states are the ones that make two other sets of states, internal and external, conditionally independent. And so it's not that a node can be even in a purely statistical sense.
So without getting into the whole, what do they physically map onto? A given node is not like a blanket state or an internal state. It's a blanket state with respect to some other states because it's part of this partitioning that's co-instantiated at the same moment. So it's an excellent point where if you know what your system of interests internal states are, then you can talk about what states blanket them with respect to some other states. But if the question is about the extended mind and what are the internal states of mind, then there's kind of this catch 22, cat in the hat scenario where if you knew where to even focus, you would have already sort of addressed the question up front. Just to sort of unpack that and connect it to some of the other ways we've talked about it.
Yeah, that's a great unpacking.
The keyword you said, and that I forgot to say, is the system you're interested in. Why I wanted to focus on this right now, because the extended mind stuff is supposed to be hyper realistic. The idea is not that, I mean, we can treat pen and paper as if they were a form of external memory.
That is something you can do, but it's not the claim that interests me, Clark, Menari. The claim is that that interests me, Clark and Menari is that pen and paper really are, with the capital A of R, really are a piece of memory in the external world. Right? They are memory in the same sense in which parts of your neural activation or memory system are memory. That is the irrelevant claim. So there is nothing observer dependent or interest dependent is really a hardcore metaphysics about how the world actually is independently about what you know about it. So that's the point that I should probably have made sooner. Well, you made it in the paper with the direction of identification runs from target variables to Markov blankets, not the other way around. So yeah, yeah. So that's the first argument.
The second one is very muddy and also very dependent on the how the leaders were on the extended mind and Markov blanket develop. But the point is basically this, that from the very beginning, that would be probably Hoi's 2016 paper, the self-evidencing brain. The message, look, there are many possible blankets and we have somehow to choose one. That sort of message got through the literature. And the way in which Yahohoi and others tried to single out the super special blankets among the many possible was to look at

self-evidencing in the long run. About keeping your keeping yourself close to your set point in the long run.
And their idea, as I understood it, was to say that, well, the machinery of the mind
is identical to the machinery that keeps yourself near your set point in the long run.
And that sounds persuasive from a free energy principle perspective,
that is not super persuasive if you are not very into the free energy principle, because there is a lot of stuff that
keep us alive and kicking in the long run. Our liver, our guts, our kidneys and prima facetion,
none of those organs does thinking in an intelligible sense.
And so the worry here that probably is never actually stated in the paper is that we are confusing,
cognizing in some broad and fuzzy sense with staying alive.
I mean, I have nothing against the claim that living things tend to cognize or tend to behave intelligently.
Planck do behave intelligently, even if very slowly, cells behave intelligently.
But I'm not super sure about equating the two and that would be probably my second point.
Okay, let's stay on this, because that's also a great point. So,
if not one in the same as staying alive what other perspectives on cognition might there be
like what is thinking other than staying alive or what is it that for example the neurons and
the glia are doing that the um cartilage in the knee is not doing well prima fascia processing
information would be a decent answer if you take DNA or at least censoring information so that that's
the question I'm struggling with right now and my two cents would be this that's we have a big problem
because not because we don't have a definition of cognition but because we have too many definition
of cognition we have too many research programs telling us what cognition is so you have ddd
classic cognitive science what is cognition well symbol manipulation then you go with the
connection is what is cognition well vector transformation sub symbolic computation then
you have the inactivists what is cognition sense making slash autonomy ddd ecological psychologist
what is cognition well interacting skillfully with affordances and so on and so forth and my big
problem is that no one here is clearly wrong no one is obviously dead wrong and that thing clearly cannot be
cognition and so what i think i would suggest is a sort of pluralism about cognition cognition
is not a single thing it's not like dancing what is dancing well moving your arms yes sometimes but
sometimes dancing might even be staying still in a certain piece of art there might be some regimented
states that's kind of ail
of us as ourselves and as people. And then there's sort of the behaviorism or at the very least
animal behavioral sciences or just behavioral studies in general, which is like from the
outside looking at a system. And so there's this gap between like us doing the thinking about
cognition and the experience of that thinking about cognition. And then the question about

recognizing cognition in other people, in children, in non-humans, et cetera. And so like,
I really like how you lay out, like there's the symbol, the sub-symbolic, inactive, and then
no one disagrees. No one is saying that bacteria make bad decisions on the whole. They're here,
aren't they? And no one's saying like a tree doesn't have a metabolic coupling with a fungus. And no one's
saying that the fungus doesn't diffuse signals, but then the aspects of these systems that people
focus their regime of attention on and where they choose to start their talks and what they decide
to talk about and what experiments they design, that frames the discussion about cognition.
That's a wonderful way to put it. And I'm a bit sad because the paper about cognition has already
been submitted. If I hadn't submitted it already, I would have stolen the framing part because it's
wonderfully said. That said, so there are two things I'd like to say about that. So the first one is
no one really disagrees. I'm not sure. I think that there is some genuine disagreement.
So the hardcore ecological psychologists, I believe, and feel free to correct me if there are any
ecological psychologists, really think that cognition is interacting with affordances,
full stop. And that there is no symbol manipulation going on in the background.
So there are genuine disagreements. What I'm not super sure about is whether those disagreements
are solvable in the sense that you can provide a master argument or even master experiments
from which you can just read off, oh, the ecological psychologist was wrong.
Oh, la, la, la. The classic cognitive this was right.
I think that the best we can do is accept some form of plurality about what cognition is.
And perhaps, but I'm not super sure, we might have to accept that there may be things that are
neither determinately cognitive, neither determinately not cognitive.
I mean, plants do behave, do cope with their environment well.
I'm not entirely comfortable about saying that there are cognizers in the exact same sense in which
I am a cognizer. Maybe it's just because their behavior is very slow if compared to mine.
Maybe it's because I have certain cultural prejudices, which I mean, they exist and
calling something a cognizer is influenced by your prejudices.
So, if you go back to the early analytic philosopher, pretty much no one thought animals think.
And now it's unbelievable that someone, for someone to have that position.
So, so there is that, but it's not super thought out.
But the big message would probably be this, put simply.
There might be something, something of a gray area between thinking with symbols,
surely cognitive, and being a rock, surely not cognitive.
The sort of big gray area in which you have intelligence, coping with the environment,
coping with the environment well, adaptability, flexibility, but still not, not be really sure
about whether there is some cognition with the capital C going on in there.
And that's the first thing.
Then you also mentioned probably about unconsciousness, but I'm very scared about that stuff.
Okay. So, another interesting piece as evidenced by the, think about what to say carefully.

Don't want to say that it's a lane we could veer off into, but right.
That is a more purely philosophical discussion.
Whereas neurobehavioral studies and cognitive science frames itself as science.
And in science, usually we think about like unique explanations, predictions, the ability to
design and control and perturb systems.
So, is this science?
What kind of science?
What are we looking for here in a cognitive model?
And how do we resolve or accommodate this tension between like unique explanations and predictions,
which is the bedrock of one type of science with the call for pluralism that you just made?
I think that the answer here is do not.
I think that the call for pluralism is not a call.
I think that the call for pluralism is not a call.
I think that the call for pluralism is not a call.
I think that cognitive science is splintered.
You have everyone doing its own thing in its own lab.
There is cooperation in the sense that I mean the various research groups are not entirely
entirely isolated. It's not the case that people working on neural neurons know nothing about
connectionism or classical computationalism or whatever. There is some communication.
But on the face of it, what I say as a philosopher looking at science, it is a splintering
of the cognitive science enterprise in a variety of groups, each with its own peculiar
research tradition. And I think that that is basically fine. If no one has a strong call or a
strong argument or a strong model that does what Newton did for physics and unifies everything
under a single banner, plurality is fine.
So there is no call. It's more something like, well, that's reality. Let's try to like it.
I'm on board with that. But with this banner,
what shape would it be? Or how would we know it when we saw it?
I have no clue. We probably just would just know it when we saw it in the sense that we would see
a research tradition starting to
absorb all others.
So we would probably see people trying to demonstrate us that there is a real and significant equivalence
between our powerful research tradition, uniform research tradition and all the others.
And probably the winning research tradition, the one that really got cognition, really got thinking,
that we would be able to make some novel prediction that no one else was able to make.
And that would be probably a strong marker that that's the way to go.
If the implicit question is, do I think that the free energy principle is that sort of research tradition?
I'm not sure.
And my answer would probably be a qualified no.

Because we do not have right now the DDD proof that is unifying as it wants to be.

It makes big unifying claims.

It makes big unifying claims.

And it might be good in the sense that they may foster cooperation at the very least.

You have at least that pragmatic advantage.

Whether it's succeeding unifying as it wants to unify is still an open question.

And so the appropriate thing to do is to, for me at least, to wait and see.

For others, perhaps, is to pick up your tools and try to unify and ensure that it really is the right thing or just argue against.

Even negative arguments are useful.

Right.

Non-action is an action.

That's true in world knowledge traditions and also in Bayesian statistics.

Like, it can be useful to wait to gather more information.

I wanted to return to what you said earlier about you need some boundaries.

And a lot of this discussion, of course, it's like extended.

How extended?

What is extending?

And what are we looking for with a boundary model in the FEP and ACTIM world?

Certainly, we talk a lot about different kinds of blankets and Bayesian graphs and things like that.

But those are not the only ways to think about boundaries.

So just wanted to learn about what are some other ways of thinking about boundaries, cognitive boundaries of different kinds?

And what are you looking for in a boundary model?

Yeah.

Not super sure about how to answer.

So my understanding is that Markov Blanket entered into debate as a sort of tool to finally understand what the boundaries of the mind are.

But there are no other big alternative if not for something said by Chalmers about action and perception be the intuitive boundaries of the mind.

But what I think he had in mind when he said that was something like, look, our intuitive image of the mind is that of something that is squeezed between perception and action.

Something like we are intuitively predisposed to say that something like the classical cognitive sandwich is true.

We encounter reward through perception and action.

So I think that's a little bit of a boundary.

And I'm not really sure whether that sort of boundary would be a boundary in the sense in which Markov Blankets

would like to be boundaries or are used to play the role of a boundary in the sense of something that
that individuates a single system and tells it apart from anything else.
The problem, I think, for the extended mind and the need for a boundary is a need fueled by the strong desire
not to say that everything counts as a cog in the cognitive machinery.
Why?
Well, because if everything counts as a cog in the cognitive machinery, then the claim that pen and paper used to do
more counts is not a big claim.
It doesn't add much, right?
In order to proclaim, well, X is part of the cognitive system to be meaningful, to be impactful.
There must be things that are not part of the cognitive system.
Otherwise, I mean, you are just claiming, well, there is this pen, which is exactly identical to any other pen.
But not so meaningful.
In the paper, you were threading together this question of short and long-term features of cognition,
like in the instantaneous action perception, cognitive behavioral timescale, and then the quote,
in the long run, where a lot of the arguments rest.
Like it reduces surprise in the long run or on average in the long run.
And then you talked about like jumping into, it would force one to count an inordinate amount of
stuff as a cog in someone's thinking machinery.
Jumping into ponds of water while on fire or shooting at a tiger to avoid becoming the tiger's
dinner both contribute to one's prolonged occupation of one's phenotypic states.
But neither ponds of water nor guns and bullets can be properly counted as cogs in the thinking machinery.
And I can totally see how one type of extended mind perspective would say like, yes, the boundaries
of the cognitive system are just so.
And they're so flexible that whatever it is in the situation that enables the cognition to happen
is the cognitive system.
At which point the lines are extremely blurred with just the system as it is.
It's like a pan-cognitivist perspective.
And that ends up, at least from a Bayesian perspective, not being informative,
because we no longer reduce our surprise by learning about anything.
And we've sort of deflated cognition away.
So where does that sit with you?
Is that a goal?
Like, what are we looking for in delineating cognitive systems from each other?
Do we prefer pan-cognitivism?
Do we prefer anti-pan-cognitivism?
Or do we want something in between where like humans are special or linguistic entities are special?
So I think that pan-cognitivism should be avoided.

In general, if you have pan-somethingism, you are on a bad road, traveling down a bad road,
because if everything counts as X, there is nothing gained from calling things X's.
No one is puzzled by things in general, I think.
So that's the first point.
Cognitivism should be avoided.
What I think is a desideratum many would buy is to have something balanced in between
individualism and the, I think, appealing intuition that sometimes interacting with stuff
at least aids cognition in a very substantial way.
So I think that the DDD go-to desideratum would be to have a bubble around DDD.
Individual, pre-theoretically understood.
Many would disagree.
Many would be comfortable with group vote, with group minds.
I think that is a minority view, but maybe they're right.
I mean, being right or wrong is not a question.
It's not a matter of numbers.
But still, to frame the debate, I wouldn't follow the minority view.
So I would keep that option in the theoretical horizon, but not use it to orient research right now.
So what I would do is to have sort of a bubble around the biological individual.
And for the timescale, I think that the right thing to do, again, just to frame things and get them started,
is to focus on short timescales.
The hyperlong evolutionary timescale
is not something that
captures our go-to intuitive sense of cognition.
And when you start framing the debate, you start from your go-to intuitive narrative standpoint.
So my sympathy is for the short timescale.
For the here and now.
It's very funny.
Like you mentioned, you know, it's not a vote.
A small group of people could be correct and a larger group could be incorrect.
Yet people often think about voting as group cognition.
Yet cognition isn't a vote.
So that's kind of an interesting tension there.
Let me ask about section five.
So it's, is vehicle externalism conditioned over Markov blankets possible?
And you said, I want to argue that resorting to Markov blankets to settle the debate over
vehicle externalism leads us to sidestep the whole debate in a very real sense,
making vehicle internalism vacuously true.
So just for like a little context, what is vehicle externalism and internalism?

Like what is the distinguishing features of these concepts?
Why are people interested in this debate?
And like, what does it matter for talking about cognitive systems?
Okay.
So, externalism and internalism are things that pop up everywhere in philosophy.
Here we go to with the vehicle variety, which is internalism or externalism about the cost
and the thinking machinery, about the material stuff that does the thinking, say.
And the, you need to have a boundary in mind to say internal and external.
And the go to boundary is the skin, basically.
The skin of the biological individual folks see understood.
Internalism would be inside the skin.
So, cognition internalism or vehicle internalism is the idea that the machinery of both is inside your skin
and more precisely inside your cranium.
Externalism is that at least some pieces of that machinery are outside.
In pen and paper, cell phones, whatever you want.
But notice, here we have an intuitive boundary, which is the skin.
So, if you go with Markov blankets, I think you end up caught in a dilemma.
Because you either start with some intuitive Markov blanket, say, at the skin, use it to define internal and external,
and then you might discover that something external to that blanket counts as part of the system.
Or you can do what, again, Jakob Hauer did, which is, well, first, I find the Markov blanket of that thing,
the real one, the Markov blanket with the capital M, and then use that blanket to define what is internal and what is external.
And if you do that, clearly, everything that matters is internal by how you have defined internal,
which is the one horn.
On the other horn, you start with the standard intuitive distinction between internal and external,
and then try to find a Markov blanket, which is fine,
but then the Markov blanket does no longer distinguish between internal and external,
or there is a tension, there are two senses of internal, and you need to disentangle the two,
the intuitive one and the Markov blanket, relative to the real Markov blanket one.
And that is not what seems to be going on in the literature, right?
So basically, that is my problem, which is a purely philosophical problem, I think,
it's a purely philosophical problem. But still, my concerns were philosophical, so it counts.
Fair enough. Is that what you were referring to when you said,
I doubt such a redefinition of internalism would buy the internalist something more than a purely verbal victory?
Yes, that's exactly the point. I mean, the internal, the vehicle internalist want,
at the end of the day, to say something like this. Thinking happens in the brain,

the internalism would be a bit more complex, but it's not a problem.
If you go on the Markov blanket path, define internal in terms of a Markov blanket,
and then discover that the Markov blanket contains the brain, but also the spinal cord,
but also my hands, but also pen and paper, but also my cell phone, and so on and so forth,
you can say, well, all that stuff is internal, so internalism is true.
But I mean, that's a verbal victory. That is not what you wanted to say at the beginning, right?
That is a form of externalism that gets passed as internalism because of how you have defined your terms.
And notice that even if you go with that definition of internalism, then there would be still, say,
super internalists that say, no, no, no, the real internal thing that does the thinking is just the brain,
and externalists and internalists or lax internalists that say, no, no, no, no,
it's the brain plus other internal stuff within that Markov blanket.
So the problem wouldn't be closed. You would still have a clash about
cerebralists and non-cerebralists, if you prefer. People that think that it's just the brain or the
brain plus the spinal cord and people that like external stuff doing part of the thinking.
It makes me think of like a thought experiment, kind of like a tracer. So like a radioactive tracer
might be injected in a blood vessel, and then you can see where it flows, or like a metabolic label
might be given to a cell, or then you can see how it gets metabolized. So the tracer, let's just say
we're following, like we're going to pluck one of the strings and trace the impact of like shining
light on the retina. Initially, it follows like nervous system paths, but then let's say that light
pattern induces the system to pick up that phone and reload a website. So now that tracer is spreading
everywhere. It's calling servers in different parts of the world, and it's like communicating through
satellites. And it's like that perturbation on the retina as space and time sort of zoom out, it's like
more and more gets wrapped into that. And so like what I'm hearing you say is if we just say, well,
wherever the tracer goes, that's internalism. Internalism is the total cognitive apparatus,
including the satellite and the server in the other country. Or one could like retreat back to the
epithelial boundary and just say like, it is extended once they are everything outside of the skin is
external. So it's sort of like, again, no one denies that somebody could see something on their retina
that causes a server far away to do something. But where is our implicit or explicit boundary?
And will we just stay with the epithelium, which gives us a boundary, but then over different time
scales, clearly different parts of that total apparatus matter. Like over one second, maybe
the oculomotor circadian and some brain region is involved with processing meaning, but then over
hundreds of years, there might be a more valid discussion about how it's actually interactions
amongst people and slower reading and writing of like books or of other resources that actually
contribute to like the semantic discussion and evolution. So it's like infuriating sometimes,
because again, no one's saying that what's on the table isn't that way. It's just like,
what are we really looking for in a cognitive theory? And what are we gaining? Or what kind of value
are we adding with these different framings?
Yeah. So again, to kick things off, things off, I would stay close to the

the basic individual. I would put the go to boundary at the scan.

And then wait and see how things evolve from that. There is an important sense, however,

in which the DDDC privilege of the scan, putting the boundary there, the scan

is inadequate.

And then wait. In the sense that really, if you look at us from a longer perspective,

perhaps as a species, you notice that

is that much of the things we do are extremely dependent on how are ail

expectation one body one mind once that expectation is in place you can discover that the mind extends

in that pen and paper and your cell phone and so on and so forth really help you think and really

should be considered part of the machinery that does your thinking if you don't have that expectation

in place it's very hard to understand what the extended mind could be for you because there is

no boundary relative to which there is an extension you can right you can't go farther farther than what

exactly you just would have the mind cognition as it is in the real world

and once you have that it makes little sense to say oh it's extended it might be extended

relative to the individual to the biological body but if you don't have expectation that the two

things go hand in hand you don't have the violation of expectation when you discover that they don't go

hand in hand you don't have the sense of discovery it's just nothing strange happening that nothing

to be amazed about nothing requiring an inquiry so that's a nice point

but as a matter of fact we have the expectation that minds are attached to individuals

so there there is a sense of wondering talking about the extended mind something that strikes us

of course a lot to say there and connect back to like what you mentioned about the cultural priors

um but sort of to hinge on the cultural priors and go in a slightly different way

how does this relate to computationalism or cognition and computers not just us using computers to have like a

human to human convo but what does computationalism add into this discussion or how would you relate

what you discussed in the paper to computational theories of cognition

i'm not super sure so i think i think i think that cognition is computational but i'm not super sure

but if asked without any reflection i would say yes of course cognition is computation

so i think i count as a computation list i see no big problem in extended computations in the sense that

if you think about computation as symbol manipulation or manipulation rule-based or rule sensitive manipulation

of certain primitives that might be symbols or sub-symbols or whatever clearly you can have rule-based

manipulation of stuff outside your body uh there is an historical point which everyone i think knows that is well originally computers were people doing the math with pen and paper uh before the the second world war before digital computer came about so there is nothing intrinsically strange about

that is what is the most extended computation and computation taking place outside your body

that being said it is true that most i would say computational models focus on the brain

and that most of cognitive scientists think that the really interesting computation happens in between the two years

and then you have i mean rodney brooks which surely is one of the inspirator of and

prime movers of embodied cognition anti-representational cognitive science

and and that really has inspired extended cognition or at least the strand of extended cognition

so you don't want to say what i would like to say is that you don't necessarily have a clash in between

computation and extended embodied cognitive science as for computational models of wider agent environment systems i

know there are on the top of my fingers or on the top of my head i don't have any examples but i can gather them if

i were to remember by my uh by my artist but there are such models and i think that's

that there should be no in principle problems of making a free energy model about agent environment system or even extended cognition

indeed i think that fristen clark kirchhoff and someone else wrote a paper about that

with a quick minimal model not sure about the title but i think is published on my language if someone wants to check

you know um

um

this is kind of uh it cracked me up with the the freudian slip with the pluralist slip you said at the top of my fingers the top of my head and it's like

aren't those the different worlds though and so it's it's just like very interesting and and it again comes back to the um the water we swim in

are we swimming the cultural priors and biases that help us like shape whether an idea is kind

of outside of the overton window like rocks can't think versus like you're a heretic if you don't

think that rocks think um as we sort of move towards a little fractal resolution of course

this will probably be a discussion for many years to come so you provided a quote from kirchhoff and

kiverstein 2019 um i'm not going to read the whole quote but it's visible on the stream and basically

they just mentioned that the markov blanket formalism as applied to systems that approximate

bayesian inference or we could even take it one more instrumental level to systems that we approximate

with bayesian inference serves as an attractive statistical framework in their case for

demarcating the boundaries of the mind but um other situations as well and then you mentioned like

perhaps this is the correct way to think about the role markov blanket should play in the debate over

vehicle externalism maybe asking where can we draw a markov blanket around the thinking machinery

yields more satisfactory results than trying to find the mark of the cognitive so um

uh this is really like a really nice point and like what kinds of questions help us move forward as we do seek like productive but also useful and accessible ways to think about these topics

what sort of question moves forward yeah i have no idea probably

the only the only sensible answer is ask questions and try to solve them some will in your retrospective

you will be able to identify the ones that really push you forward from the ones that were not well posed

so probably the only sensible question to which the only sensible answer is wait and see

uh as for ddd kirchhoff and kievstein quotes uh i'm not super

so

a big problem i had with markup blankets and extended mine is this is to me at least not super clear whether when markup blankets are used the markup blanket is something out there in the world like some mathematical entity instantiated somehow in the world or whether they are some sort of accurate representation of what is going on or whether they are just a mathematical tool to make sense of what is going on it seems to me that those three things are used together uh when it comes to discussing the boundaries of the mind but there are three different things i mean there are

three different ideas about what the markup blankets are and by adhering to one coherently you have a very different outlook and a very different metaphysics say of the free energy principle if markup blankets are things out there in the real world then it seems that you must take the free energy story at face value

you must be an hardcore realist about the free energy principle if they are just a tool you can take the free energy principle to be just perhaps a sort of mathematical machinery you can deploy to build interesting models of phenomena you want to study you can try just as you can use standard computational theory to build models of things you can use the free energy principle or you can use dynamical system theory you have various tools you can deploy

you can use the free energy principle or you can use the free energy principle or you can use the technology to be a tool you can use the free energy principle or you can use the free energy principle

but the point is that if you go super instrumentalist oh that's just a model

and i do what i want with it my only guide is not truth but

is not to make states

But you also have a very straight outlook on reality, I think,

with these sort of fractal boundaries within boundaries
that are reciprocal and so on and so forth.
My two cents would be that to follow the pragmatic,
non-realist usage of the free energy principle,
to take it as a model building tool.
But don't trust me.
There are my two cents and I'm not super savvy
about instrumentalism and model building.
Definitely this question or tension of instrumentalism
and realism kind of map and territory in Mel Andrews 21 paper and several other times we've
seen this discussion arise, of course, is active inference just a statistical model that we can
apply to action cognition perception, just like we could apply a linear model to height and weight,
but that doesn't make height and weight a linear regression. So does applying the active
inference structure to neurobehavioral data that cannot be taken as positive evidence that that
system, for example, is doing active inference. So that's definitely a point well taken. And the
pragmatic turn, which is associated in cognitive science with a movement towards action orientation,
and then there's like this meta pragmatic turn in cognitive science, which is like
better theories are more useful. And that actually is in contrast with what we were speaking about
earlier with science as providing explanations, predictions, because the pragmatism actually
highlights the ability to design and control. It's like, well, if I used the epicycles to
explain, predict, design, control, it's useful enough. And that in a pragmatic world would be sufficient.
But then somebody is always coming out of left field or right field or whatever, with the questions
about how things actually are and saying, well, but they're not actually moving in that way.

Yeah, I perhaps do not see any strong clash between pragmatism and irrealism. What you definitely have here are two, say,

tendencies, one to be hyperplotonic and say, look, reality is this way. There is nothing in your data that can prove me wrong.

which is even respectable sometimes for certain class of entities like mathematical entities and the opposite tendency to say,

I don't really care about how things are. I just care about prediction and control and making good science and predicting stuff and making discoveries.

I think that a sensible observation is that typically theories and models that tends to make discovery and give you good prediction have a degree of accuracy, tend to capture something about how things are.

I mean, you cannot proceed arbitrarily in model building and do whatever you like.

You have some sort of resistance and intuitively I'd say that resistance is reality.

And so even going fully pragmatism, prediction and control only, you get something about reality.

Maybe what is lacking is the sort of platonic ideal reality that is super independent about what you know.

But I mean, that is something you encounter rarely and only when you do philosophy.

So if it gets left behind, it might not be the biggest tragedy of the human intellectual history.

Let's put it in this way.

So one thought is that in active inference, the way that we think about action selection, policy selection is as related to providing pragmatic value, which like brings you into alignment with your preferences and then epistemic value, which reduces your uncertainty.

And it's almost like the utility is loaded on the pragmatic side.

But of course, just narrowly pursuing the pragmatic value ends up leading into a blind corner.

And so it's good to balance both and to have like diversity of thought within a person within a scientific community.

We don't need to be 100 percent convinced or only on one side of the line as individuals or in a collaboration.

And then to connect that to one other point that you made in the paper.

So it's on.

Not sure if there's page numbers, but it's there should be page numbers.

But there are.

But there aren't.

So I think it's the online version.

And so without page numbers.

Yeah, exactly.

So you were talking about benchmarks that adjudicate whether something counts as a constituent of the thinking machinery.

And so that's like something we've been talking about.

So then you wrote traditionally this benchmark is provided by the mark of the cognitive that one endorses.

Again, relate to what we've been discussing.

That is by what one implicitly or explicitly takes to be necessary or sufficient to make something a genuine contributor to thinking in the broadest sense.

So that I'm thinking about this paper.

So it's a single author paper.

However, you mentioned like in a footnote, like this observation and the example are due to Chris DeLega.

You mentioned in the acknowledgements.

There's some people's names.

There's citations, which is, of course, in science, how we communicate like attribution of ideas.

And then there's even like a little discussion with a reviewer.

And so it's like there's all these different contributions actually to this cognitive artifact, this epistemic artifact.

And so it's like you're supporting the biological individual epithelial boundary, single author paper, all these external contributors.

Like they can get mentioned, but they're not authors.

And then like if it were going to be written by the radical pan cognitivist, like why not make everyone an author?

Just let's have everybody as a contributor.

And so just sort of like a funny little microcosm there.

No, no one, not even the hyper radical would put everyone as an author because no one likes the academic .

No, I tend to preview that the individual boundary, but I mean, I am pro extension at the end of the day.

I think I mentioned it in the paper that I am in favor of cognitive extension.

And perhaps because I want cognitive extension to be real, I favor some conservative boundary, which is easily transpassed.

This is perhaps more psychoanalysis about my choices and my preferences.

But yes, I mean, I'm on board with GTA that you can have cognitive processes that are that run through multiple brains.

I'm not super sure what how common they are, because even unless you do some real extreme brainstorm.

So typically the input coming from others is very limited or not very sizable.

But, I mean, there are, I would say,

generally multibrain cognitive processes.

Like a conversation.

Yeah, like very deep conversations

in which at the end of the day you agree.

You do bring about some well-defined artifact or output.

Just free-floating conversation,

I wouldn't say that is a genuinely extended cognitive process.

But, if you want to know,

there is a paper by Jelle Brunneberg and Regina Fabre

in which they take all the extended mind stuff

with mind wandering.

And that is fascinating because mind wandering

is clearly a cognitive process,

is clearly thought with the capital T,

I would even say,

but is not something that has a definite output.

There is no set endpoint you might want, right?

And yet they claim it might extend.

And if you take this sort of no set output,

no set endpoint perspective,

then maybe even causal conversation may qualify

as intrapersonal cognitive processes.

But I'm not super sure about how one could develop that idea precisely.

It kind of makes me think about,

in a very transposed,

analogical way,

like to integrated information theory.

Like if two people are speaking at each other,
but they're both having the conversation
that they would have with a wall,
then it's like they're speaking past each other.
It's not like an emergent conversation.
Whereas if people are engaged in like active listening
and like actually integrating some novel stimuli
and then coming up with something
they wouldn't have said to a wall,
then that is like where that third space arises,
where things can be productive.
But then also that's very interesting
that they might not be.
And one other note on the mind wandering
is as our technology in the niche changes
with some neural recording opportunity,
say, just let your mind wander.
And all of a sudden through biofeedback
or the enabling of new cognitive linkages,
that mind wandering actually could leave a trace
on the environment in a way where
if it's just brain bound, it wouldn't.
Which I think returns us to that earlier question,
just like that one thing I want to address
about like composition of cognitive systems.
So whether we're internalist or externalist or whatever,
how do we think about this space
of the adjacent possible for cognitive systems?
Like what can we think of next?
What can we think of next?
Or how could we think differently
if there were some structural changes to our environment?
Like the pen until it existed
or the writing implement until it existed wasn't there.
And there was pre-adaptations
and the evolutionary basis for utilizing it, et cetera.
But when it was there,
it changed the game in a quantum way.

So how do we think about those kinds
of transformations of cognition?
I'm super not an expert about this topic.
And I'm a bit ashamed by this, frankly.
But I, and I need to catch up.
But I think that the probably strongest case study
you could make for transformation
is with reading and doing math.
We have, of course, basic visual capacities,
but reading seems to be a purely cultural thing,
something that you need culture to learn
and something that deeply transforms
your cognitive abilities.
So just the ability to target your own thought crystallized
in a written form gives you a level of meta-reflection
that would be hard to maintain with your brain alone.
Or the ability to crystallize your thoughts
in a scrap of paper,
leave it there to ferment,
and then go back at it one week later,
two weeks later, three weeks later,
with a fresh mind.
This sort of thing is not something you can do
without externalizing your thoughts.
You can just remember your past thoughts
and objectively look at them
as you objectively look at your past-fold written paper.
And the other thing, of course, is math.
We have some basic familiarity with numbers,
that is, biologically endowed,
but pretty much from arithmetic on,
it is all bootstrapped by culture.
And, of course, here you have the biggest transformation possible
with mathematical objects,
modeling, and all sorts of things.
So those would probably be the case studies
to try and have a sense of what DDD,
space of a decent possible mind, might be.

And this is pretty backward-looking.
I'm trying really hard right now
to think something more forward-looking,
something more propositive,
something more hands-on, let's build a new mind,
but I don't have any idea right now about that.
Well, look back to look forward.
It's an interesting idea and theme.
I, of course, wish you luck on the forward-looking,
but when you talked about asking questions,
it made me write down and subsequently remind myself later,
either the question will have been asked already,
doesn't mean that it's been satisfactorily answered
to anyone or to you,
but either the question has been asked
in its rhetorical structure,
and then it'd be awesome to point people
towards a knowledge resource,
or it's a novel question,
and novel questions are awesome.
And then looking back,
it may be possible to know
which of those novel questions
led to a useful line of discussion or application,
but to wait before the question
and say, well, I only want to ask the best question,
it's like, in a way, we don't know
since it's so many steps down the road from the question,
at least one or two,
till we can evaluate the impact of the question.
And so, like, as a hack or a heuristic,
having rigorous scholarship
so that when the questions have been asked,
we're not doing truly redundant work,
but then also, like, recognizing and raising up new questions
helps expand the frontier and the fringe,
and some fraction of those may develop into useful avenues.
And then later on, people might say,

well, it was obvious.
Somebody should have connected the sequential database
with the encryption and with an economic system,
but those things existed long ago,
and it's like looking back
that some of the syntheses conceptually
or technologically post-hoc make more sense.
But again, waiting and not acting
for it to make sense a priori,
it's like it doesn't seem like just a 2022 thing
that that's not going to work.
That seems to be, like, in principle,
a very challenging way to expect effective action.
Yeah, I'm not super sure I got the point.
So, GDR would be this,
that you don't have any sort of precognition
about where a question will lead you.
You cannot evaluate questions beforehand
before asking them to see,
oh, this question right here,
a long way to go, it's a good one,
and this other is a waste of time, right?
Is that the point?
Yeah, even if there was no,
yeah, we shouldn't prejudge questions.
We could easily sort questions
into those that have already been asked
and those that have not been asked.
And so to those that have been asked,
we can use scholarship.
To those that have been asked,
we can, like, recognize novelty.
So even we don't need to finesse that too much.
And just to kind of connect that,
however vaguely,
to your idea of, like, looking back,
but then the challenge of looking forward
in cognitive sciences sometimes.

Yeah, it's truly a challenge,
but I think it's a big challenge for me
because I'm illiquid to look forward
in cognitive science.
If you probably ask people that do
build cognitive architectures
and cognitive models for a living,
I think they would be
much more forward-looking than they.
I mean,
you have something you'd like to try,
some twist on the architecture
you'd like to put,
and that is, I think, forward-looking, right?
Well, what is your forward-looking,
like, over whatever time horizon
or capacity set that makes sense?
Like, what kinds of questions
or experiments or research areas
are you, like, excited to build on
from this paper?
I'm not super sure
where this paper is headed.
I mean,
I'll continue probably
to work on extended minds.
I still think
I'll be working on
predictive processing
in a sort of very
vapid fashion.
Probably I'll try to merge T2
in a more positive way,
probably,
but I'm not super sure,
probably in computational terms,
but this is super,
the super,

what I want to do
with my postdoc,
if I had one.
Nice.
It's not a concrete proposal
right now.
Hypothetical postdocs
are the best postdocs.
It's a list of wishes.
What I wish to do,
well, I wish to do,
yeah, probably that.
That would be one thing.
And another thing
I'd like to do
is to try
and
say
open up
and specify
the grip metaphor.
You have Andy Clark
and
those other people
telling you
what predictive processing
slash the free energy principle
gives you
is a grip
on the affordances
that matter.
It's not super clear
what the grip is
apart from
not stumbling
into things
and moving around
appropriately.

the epistemic
sort of contact
with the war
is left,
I think,
unanalyzed.
But I have no
big idea
on how to analyze it.
It makes me think
about how
words related
to our physical
cognition
end up
describing our
interior landscape
like
get a grip,
you know,
can you see
what I'm
thinking about
or let me
sketch it out
for you.
Comprehension
means to grasp
with a hand.
So like
there's so many
and of course
the whole
optimal grip
active inference
as getting grip
on things
from a physical

perspective
like as the hand
goes for the
cup of coffee
but then
okay,
now we transpose
that into
the cognitive
space
and we
in a conversational
way do talk
about grip
and then
here's all
of these
formalisms
that it can be
hard to get a grip
on
but are they
about grip
or like
what would that
actually mean?
That's a very
interesting question.
Yeah,
I mean
there is
a way to
impact
the grip
idea in terms
of
Bayesian
inference

but that
is a bit
too internalistic
for my
liking
the idea
that perception
is a form
of inference
that you are
inferring
every time
everything
it works
in the model
domain
the model
is
inferential
whether
the thing
itself
is
inferential
I'm not
super sure
and I'm
I'm not
sold on
the idea
that the
inferential
talk
really captures
what goes
on in
most instances
of cognition

I mean
most
of our
thought
is not
described
inferentially
when I
let my
mind go
and start
mind wandering
not inferring
things at
random
I'm mind
wondering
when I
draw an
inference
I draw
an inference
typically
with
high
concentration
to one
specific
topic
and usually
the
date of
pen and
paper
so the
inferential
talk might
not be

optimal
but still
I feel
the need
to do
to give
some sort
of epistemic
connotation
to the
grip
is not
just
not
stumbling
into
things
and moving
around
quickly
and swiftly
there is
some sort
of comprehension
and intelligence
and mindfulness
into that
that going
with a
pure
good behavior
good behavior
of patterns
vocabulary
might not
just miss
but it
might even

let you open
to the charge
of being a
behaviorist
of being
a philosophical
zombie
without any
inner life
and that sort
of things
we need
some mentalistic
lexicon
some epistemically
charged lexicon
which is not
full-grown
decart
and a major
challenge
would be
to develop
it
it makes
me think
about like
an ocean
and the river
and then
where they
meet
and it's
brackish
cognition
and pure
cognitive
the rest

cognitive
and then
there's
the
physicalism
and
we have
the ability
to like
be uncertain
actually
and to
evaluate
and to
oscillate
between
different
perspectives
or different
regimes of
attention
on these
different
ideas
like
more
highlighting
the physical
boundaries
of a system
more
highlighting
the cognitive
more
highlighting
the statistical
causal
which is

where the
Markov
blanket
concept
comes from
from
pearl
and
bayesian
causal
graphs
and those
causal
graphs
the nodes
and the
edges
may or may
not
link
to anything
that is
a physical
thing
or a
cognitive
thing
so it's
a very
open
and
interesting
area
is there
anything else
that you
like wanted
to just

bring up
talk about
ask
no I
don't
think
so I
think it's
now time
to thank
you for
the space
the opportunity
and the
wonderful
talk
to thank
the
audience
which I
have not
seen
because I
can only
see you
apparently
from my
screen
but hi
audience
I hope
you enjoyed
it
I hope
you didn't
find me
too rambly
if they're

listening this
far
then I
don't think
either of
us are
too rambly
for them
oh maybe
they're just
doing exercise
driving around
and once
I'm
standing
back
around
we've acted
in France
awesome
well yeah
Marco
thanks for
the paper
and the
contribution
and for
joining for
a guest
stream
you're always
welcome to
continue the
discussion
and also
thank you
for all
of those

who
listened
and paid
attention
for this
conversation
great
okay
see you
later
see you
later
bye
everyone